

SLED Datacenter Market Intelligence: A Natural Language Processing Approach to Transforming Public Procurement Data into Actionable Sales Intelligence

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I. Executive Summary

The State, Local Government, and Education (SLED) sector represents a structurally distinct segment of the information technology market. Characterized by rigid multi-year budgetary cycles, bureaucratic procurement processes, and non-standardized data practices, the SLED market poses significant visibility challenges for technology vendors seeking to engage prospective customers at the right moment and with relevant, pre-qualified intelligence. For Nutanix, a leading provider of hybrid multicloud infrastructure solutions, the absence of granular, reliable intelligence regarding existing customer technology investments, procurement timelines, and reseller relationships impedes the effectiveness of pre-sales engagement and strategic resource allocation.

This project addresses that gap by designing and implementing an end-to-end analytics pipeline that transforms publicly available government procurement records into structured, actionable market intelligence. The primary data source is GovSpend.com, a commercial aggregator of public-sector purchase order data. The raw dataset comprised approximately 2.1 million line-item records spanning five years of procurement activity across SLED entities in the southeastern United States. Because procurement descriptions are entered as free-form text with no standardized vocabulary, the raw data is largely uninterpretable at scale without automated processing.

To address this challenge, a multi-stage Natural Language Processing (NLP) pipeline was constructed in Python. The pipeline employs a logistic regression “gatekeeper” model, trained on a manually labeled sample of 500 records, to filter non-IT transactions from the dataset. A second stage uses the spaCy NLP library with a custom named entity recognition ruleset to extract technology vendor identities and product categories from each qualifying record. The combined pipeline achieved 90% accuracy on vendor extraction when evaluated against a hand-labeled “Golden Dataset” of 500 records. The processed output was used to populate an interactive Microsoft Power BI dashboard providing Map, Customer, and Reseller views, and to generate a secondary industry trend report analyzing on-premises versus cloud spending patterns across three SLED sub-segments: Higher Education, State Government, and K–12 Education.

The resulting tools enable Nutanix pre-sales engineers to rapidly identify prospect technology profiles, procurement patterns, and reseller relationships—intelligence that was previously inaccessible due to the

unwieldy nature of existing data platforms. The approach demonstrates the practical application of NLP, supervised machine learning, and interactive data visualization in a real-world commercial analytics context.

II. Introduction and Project Purpose

2.1 Business Problem

Nutanix's Pre-Sales organization is responsible for enabling its Systems Engineering resources with the customer intelligence necessary to conduct high-value consultative engagements. An informed pre-sales interaction—one in which the engineer arrives with knowledge of a prospect's existing technology platforms, data center investments, and application portfolio—elevates the conversation beyond basic discovery and allows for substantive discussion of integration pathways, product compatibility, and strategic alignment. A core engagement vehicle is the Technical Check-In, in which a Systems Engineer reviews a customer's current Nutanix environment and broader data center objectives.

In practice, however, obtaining a comprehensive view of a prospect's technology environment is constrained by two factors. First, client-facing time is a scarce resource, and engineers cannot always afford to spend it on basic data-gathering that could be performed in advance. Second, clients are often reluctant to disclose detailed infrastructure information until trust has been established over time. These constraints create a clear demand for a pre-built, continuously available intelligence resource that surfaces relevant data automatically before a customer interaction occurs.

The proposed solution is a self-service market intelligence tool that provides Nutanix engineers with rapid visibility into a prospect's technology spend by category (Compute, Storage, Private Cloud, Public Cloud), by technology brand, and by reseller channel—all derived from publicly available procurement records and delivered through an accessible, interactive interface.

2.2 Project Scope and Objectives

The project scope encompassed five years of purchase history for public-sector entities within Nutanix's SLED sales territory in the southeastern United States, with a particular focus on the states of Georgia and Florida. After applying geographic, entity-type, and temporal filters using the GovSpend platform, the resulting dataset exceeded two million rows—a volume that surpassed the platform's online processing capacity and required a manual bulk export for local analysis.

The project had three primary analytical objectives: (1) to classify each procurement record as either a genuine datacenter IT purchase or a non-IT transaction, thereby removing noise introduced by the limitations of GovSpend's native UNSPSC-based categorization; (2) to assign each qualifying record to one of several datacenter technology categories (e.g., Compute, Storage, Networking, Security, Backup, Virtualization, Public Cloud); and (3) to extract the technology vendor or brand name associated with each purchase. Together, these three outputs enable customer-level analysis of procurement history, competitive install-base identification, and reseller channel mapping.

III. Methodology and Approach

3.1 Data Source and Acquisition

The primary data source was GovSpend.com, a commercial provider that aggregates publicly available procurement information from government-funded entities through a proprietary combination of public records requests and automated data extraction (GovSpend, 2024). Nutanix holds an active subscription to the platform, and all data usage fell within the terms of the applicable agreement. The end product is intended for internal use only. The dataset contains no personally identifiable information (PII), and no HIPAA or other privacy-sensitive data categories are implicated.

The native data schema includes fields for the procuring agency, the reseller, geographic identifiers, purchase date, line item description, unit price, and GovSpend’s proprietary implementation of the United Nations Standard Products and Services Code (UNSPSC) classifications (UNDP, 2024). Inspection of the UNSPSC classifications revealed frequent and systematic misclassification—particularly for technology purchases—rendering the native categorization insufficient for this analysis. The core analytical challenge was therefore the “Line Item Description” field: an entirely free-form text entry with no enforced vocabulary, format, or controlled terminology. Descriptions range from detailed technical specifications to cryptic abbreviations, requiring a robust automated processing approach.

3.2 Processing Pipeline Overview

The processing pipeline was designed as a sequential, four-phase funnel that progressively reduces and enriches the raw dataset. All development was performed in Python 3.12 using Visual Studio Code. Core libraries included pandas for data manipulation and ETL operations, scikit-learn for machine learning model training and evaluation, and spaCy for NLP-based entity extraction. Microsoft Power BI was used for final visualization and dashboard delivery.

3.3 Phase 1 — Data Acquisition and Preliminary Filtering

The first phase involved filtering the GovSpend platform for the target geographies, entity types, and a five-year historical window. The resulting dataset of over two million records was exported in bulk CSV format due to platform limitations at that scale. A preliminary UNSPSC-based filter was applied to narrow the dataset toward information technology categories; however, this filter was found to introduce significant leakage of non-IT transactions and could not be relied upon in isolation.

3.4 Phase 2 — Datacenter Classification via Logistic Regression

To address the misclassification problem, a binary logistic regression classifier was trained to distinguish genuine datacenter IT purchases from non-IT transactions. A random sample of 500 records was drawn from the filtered dataset and manually reviewed and labeled by the analyst as either “datacenter” or “not datacenter.” This labeled dataset served as the training corpus for what was termed the “gatekeeper” model.

The model incorporated two feature categories: text features derived from the “Line Item Description” field using Term Frequency–Inverse Document Frequency (TF-IDF) vectorization, and a numeric feature representing the unit price of each line item. TF-IDF was selected for its effectiveness in identifying discriminative terminology in short, domain-specific text without requiring large training corpora. Rows

subsequently classified as “not datacenter” were excluded from further processing, reducing the dataset to approximately 1.9 million records. Model performance metrics are presented in Figure 1 below.

--- NEW MULTIMODAL EVALUATION REPORT ---				
	precision	recall	f1-score	support
Not Datacenter (0)	0.97	0.96	0.96	1997
Datacenter (1)	0.98	0.99	0.99	5647
accuracy			0.98	7644
macro avg	0.98	0.97	0.97	7644
weighted avg	0.98	0.98	0.98	7644

Figure 1. Gatekeeper Model Performance Metrics (Logistic Regression with TF-IDF and Unit Price Features).

3.5 Phase 3 — Vendor and Category Extraction via Named Entity Recognition

Following the gatekeeper filtering step, each surviving record was passed through a Named Entity Recognition (NER) pipeline designed to extract two key features: the technology vendor or brand (e.g., Dell, Cisco, VMware, Nutanix, AWS) and the datacenter technology category (e.g., Compute, Storage, Networking, Security, Backup Software, Virtualization, Public Cloud). The NER pipeline was implemented using spaCy, a production-grade Python NLP library well-suited to custom entity recognition tasks.

Because procurement descriptions do not conform to any particular standard format, a pre-trained general-purpose language model would not reliably identify technology brand names in this context. Instead, a custom entity_ruler component was created within a blank spaCy pipeline and populated with a manually curated dictionary of technology brand names, product families, and category keywords drawn from industry research. The dictionary was structured using spaCy’s pattern-matching format, enabling both exact-match and token-sequence matching across varied description formats and abbreviation styles.

To evaluate and iteratively improve extraction accuracy, a “Golden Dataset” of 500 manually annotated records was constructed with ground-truth labels for vendor and category. The spaCy pipeline’s extractions were scored against these labels through multiple rounds of review. In each iteration, the entity dictionary was refined to address false positives, false negatives, and ambiguous or multi-vendor descriptions. Following iterative tuning, the pipeline achieved 90% accuracy on vendor entity extraction against the Golden Dataset. Where vendor extraction was not possible due to insufficient description content, records were still retained and labeled at the datacenter category level, preserving analytical value even in the absence of brand-level resolution. (See Figure 2 for graphical concept).

Figure 2: NLP Data Pipeline

DataCenter Spend

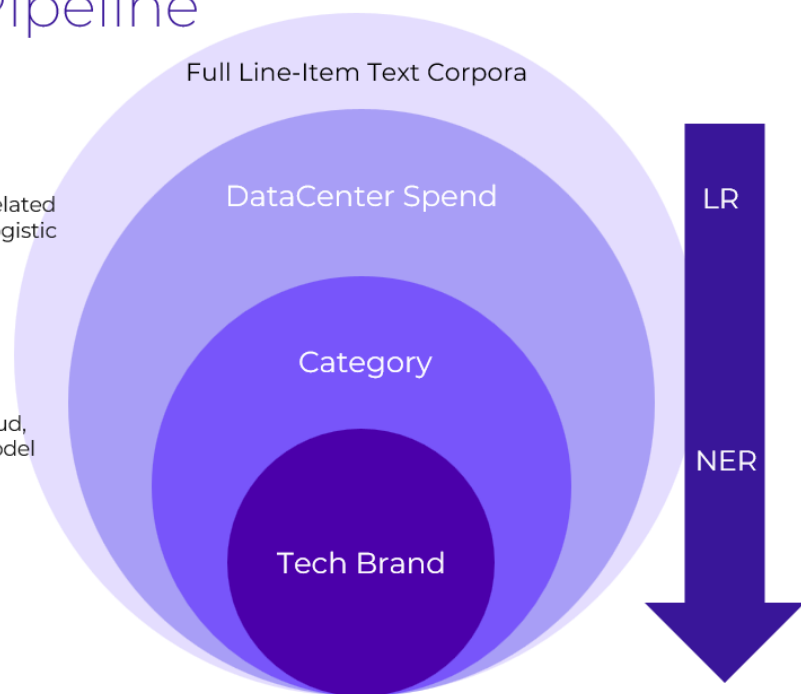
Classify the Line Item Purchase as Either a Datacenter related Purchase, or Non-Datacenter related purchase with a Logistic Regression model (LR).

Category

Extract and detect features such as Compute, Storage, Networking, Security, Backup, Virtualization, Private Cloud, Public Cloud, SaaS, with a Named Entity Recognition Model (NER).

Vendor/Brand

Represents Company Name. For example, AWS, Cisco, VMWare, Nutanix, also with NER.



IV. Implementation and Technical Infrastructure

Following successful validation of both the gatekeeper model and the spaCy NER pipeline on a development dataset of 38,000 rows, the full processing pipeline was executed against the complete 2.1-million-row dataset. The available compute environment was limited to a single client laptop equipped with 32 GB of RAM and an 8-core CPU. To accommodate this constraint without triggering out-of-memory errors, a chunked processing strategy was implemented: the source data was loaded, processed, and written to disk in sequential batches of 100,000 rows, with RAM explicitly cleared between each batch using Python's garbage collection utilities. This approach enabled full-dataset processing without hardware upgrades or cloud compute resources.

The output of the pipeline was a structured CSV file optimized for ingestion into Microsoft Power BI. Each row in the output file represents a single procurement line item and includes the original agency, reseller, geography, date, and price fields alongside the newly assigned datacenter category, extracted vendor identity, and a master category grouping field. This file served as the sole data source for the interactive dashboard delivered as the primary project output.

V. Results and Deliverables

5.1 Primary Deliverable: Interactive Power BI Market Intelligence Dashboard

The primary project deliverable is an interactive Power BI dashboard providing three distinct analytical views of the processed procurement data: the Map View, the Customer View, and the Reseller View. Each view is designed to serve a specific pre-sales intelligence use case and is accessible to Nutanix sales and engineering personnel through the organization's existing Microsoft 365 service subscription that allows for individual use.

5.1.1 Map View

The Map View provides a geographic overview of technology spend across the territory. Users can navigate the interactive map and click on individual agency markers to reveal a pie chart summarizing that entity's total technology spend breakdown by category (Figure 3). An accompanying interactive bar chart displays total procurement revenue by technology vendor, and selecting a vendor filters the accompanying data table to show agency name, technology brand, reseller, and total spend for matching records. This view enables rapid territory-level scanning and identification of high-value prospect clusters.

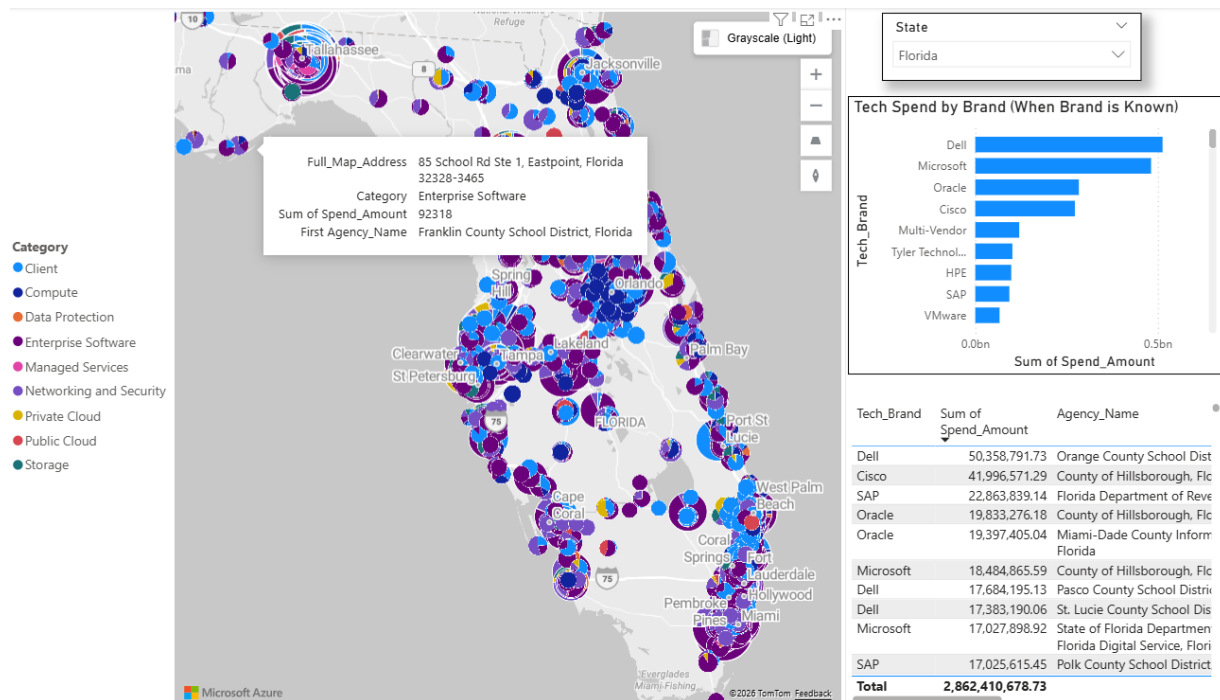


Figure 3. Power BI Dashboard — Map View. Geographic distribution of SLED technology procurement by category and vendor.

5.1.2 Customer View

The Customer View is the central intelligence interface for individual account research (Figure 4). A user selects a state and an agency name, and the view populates a stacked area chart displaying that agency's total IT procurement spend by technology category across the five-year historical window. This timeline visualization reveals budget cycle patterns, technology refresh cycles, and category-level investment trends. Because IT asset investments in the public sector typically operate on three- or five-year renewal cycles, this view provides actionable timing intelligence that can be used to anticipate upcoming procurement decisions. The view also identifies which resellers the agency has historically transacted through, supporting informed partner engagement strategy.

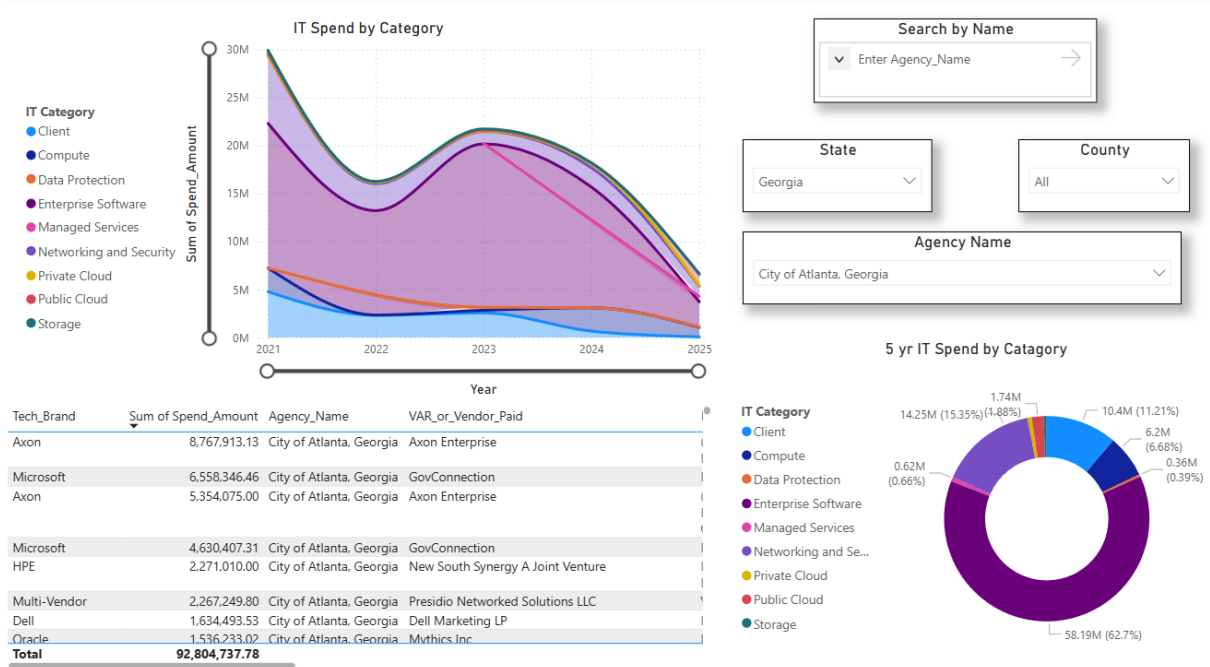


Figure 4. Power BI Dashboard — Customer View. Five-year IT procurement timeline for a selected agency, segmented by technology category.

5.1.3 Reseller View

The Reseller View mirrors the structural design of the Customer View but reorients the aggregation around reseller entities rather than end-customer agencies (Figure 5). Selecting a reseller populates a view showing the total sales volume that organization has transacted through SLED procurement channels, broken down by technology brand. This provides Nutanix sales leadership with a concise portrait of a reseller's business profile in the territory—which technology brands they most frequently represent, which customer entities they serve, and the overall volume of their SLED activity. This intelligence supports more strategic partner planning and reseller engagement conversations.

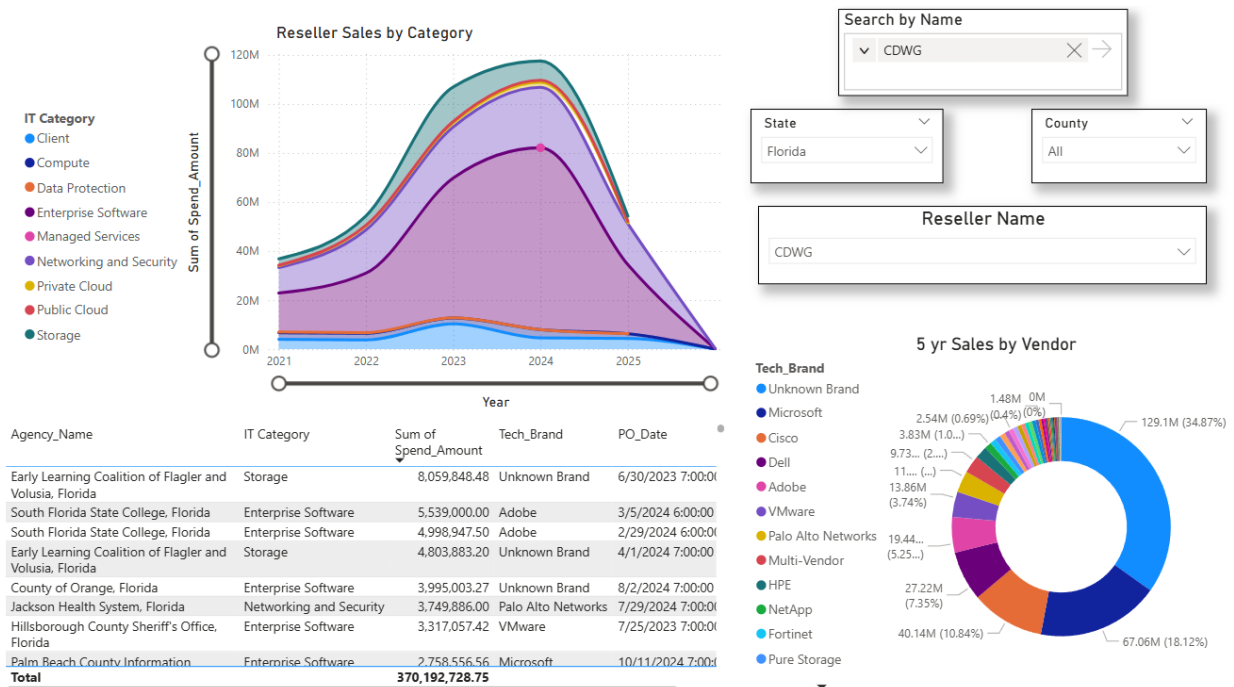


Figure 5. Power BI Dashboard — Reseller View. Reseller-level sales volume and brand distribution across the SLED territory.

5.2 Secondary Deliverable: SLED IT Spending Trend Analysis

A secondary deliverable consists of a market trend analysis examining on-premises versus public cloud spending patterns across three SLED sub-segments over the five-year observation window. This analysis surfaces broader strategic insights regarding infrastructure investment trajectories within the public sector and provides additional context for Nutanix’s product positioning in the SLED market. *Note: The year 2025 data was excluded for trend analysis due to the completeness of the data in from the source. Because GovSpend utilizes public records requests, and each jurisdiction has different laws that determine how long they have to publish data, there is a known lag between the date of purchase and the time the data is collected and published.*

5.2.1 Higher Education

Analysis of 116 Higher Education institutions reveals that on-premises datacenter spending—encompassing hardware and infrastructure software—remains substantially higher than public cloud expenditure in absolute terms (see Figure 6). However, the trajectory of on-premises spending has been comparatively flat over the most recent four years of the observation period, while cloud expenditure shows a modest upward trend. This pattern may reflect the accelerating adoption of Software-as-a-Service (SaaS) applications in higher education, which reduces pressure for incremental on-premises capital investment without yet displacing the existing install base.

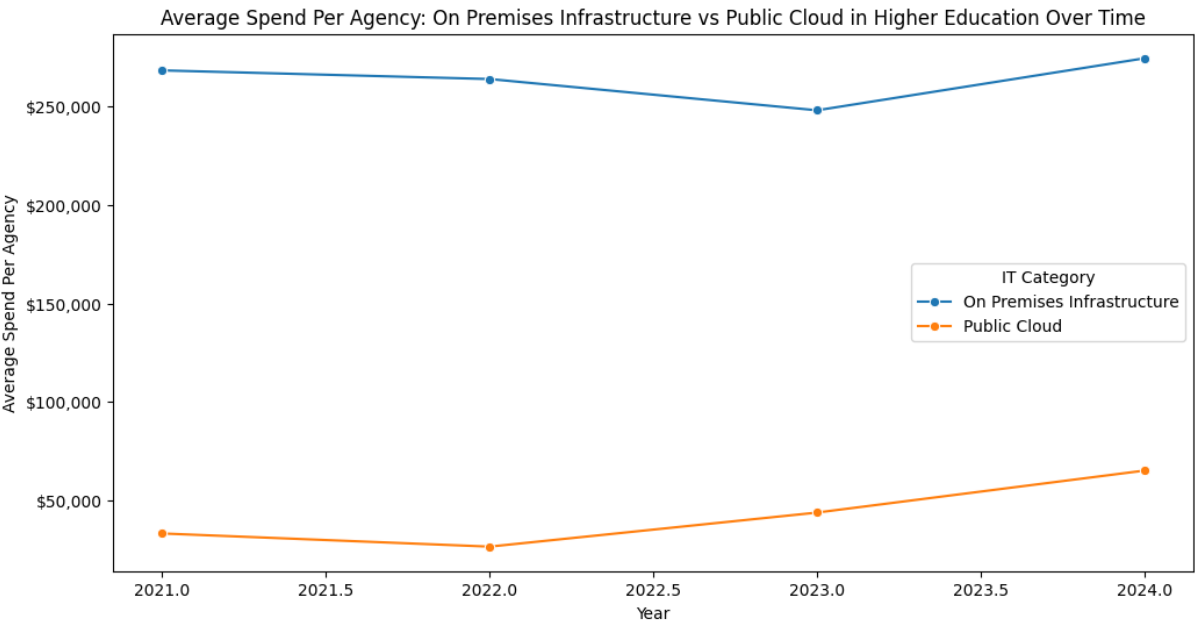


Figure 6. On-Premises vs. Public Cloud Spending Trends — Higher Education (n = 116 institutions).

5.2.2 State Government

The State Government sub-segment, derived from a sample of 286 agencies, presents a different pattern: both on-premises and public cloud spending are growing year-over-year across the observation period (Figure 7). This concurrent growth suggests that State Government entities are expanding their overall technology budgets and adopting cloud services as a complement to, rather than a substitute for, existing

on-premises infrastructure. For Nutanix—whose hybrid multicloud product portfolio is positioned at this precise intersection—this dynamic represents a favorable market condition.

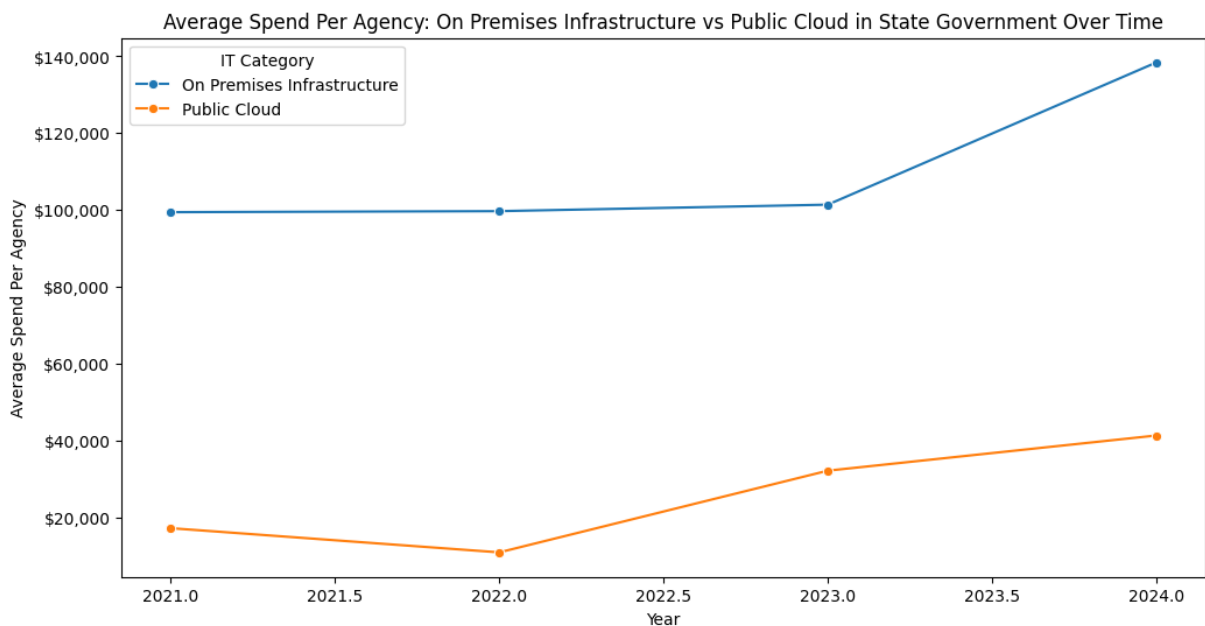


Figure 7. On-Premises vs. Public Cloud Spending Trends — State Government (n = 286 agencies).

5.2.3 K-12 Education

The K-12 sub-segment, analyzed across 2,195 entities, displays the clearest cloud migration trajectory of the three groups: on-premises spending is declining over the observation period while public cloud expenditure is increasing (Figure 8). This pattern is consistent with a structural shift in K-12 technology strategy toward cloud-native platforms and managed services, which reduce the operational burden on resource-constrained district IT departments. For Nutanix, this trend highlights the K-12 segment as one requiring differentiated positioning relative to the State Government and Higher Education sub-segments.

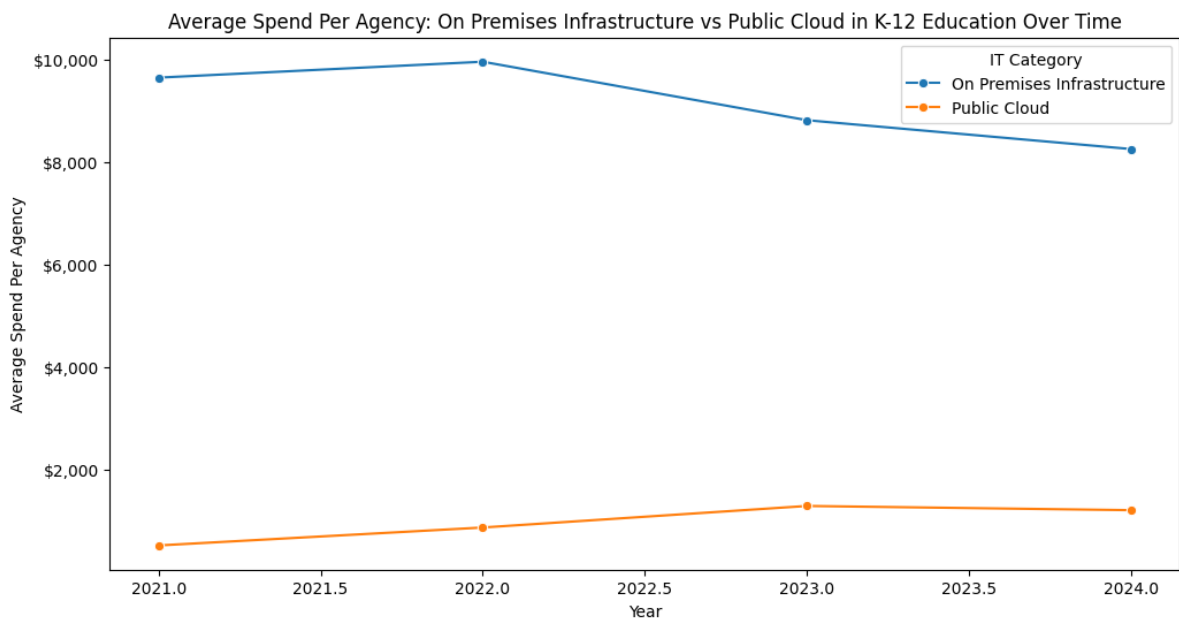


Figure 8. On-Premises vs. Public Cloud Spending Trends — K-12 Education (n = 2,195 entities).

5.2.4 Competitive Intelligence: VMware by Broadcom

The dataset also enables direct competitive analysis at the vendor level. As Nutanix’s primary competitor in the virtualization and private cloud market, VMware by Broadcom represents a key benchmark (Figure 9). Procurement data for the same sampled Public Agencies over 4 years reveals that SLED Agencies VMWare spending peaked in 2022 and has since experienced two consecutive years of decline. This pattern aligns with the timing of Broadcom’s acquisition of VMware and the subsequent restructuring of VMware’s licensing and pricing model, which de-emphasized the public sector in favor of large enterprise accounts. For Nutanix, this competitive dynamic represents a meaningful opportunity in the SLED segment, as public-sector entities facing VMware price increases may be actively evaluating alternative platforms.

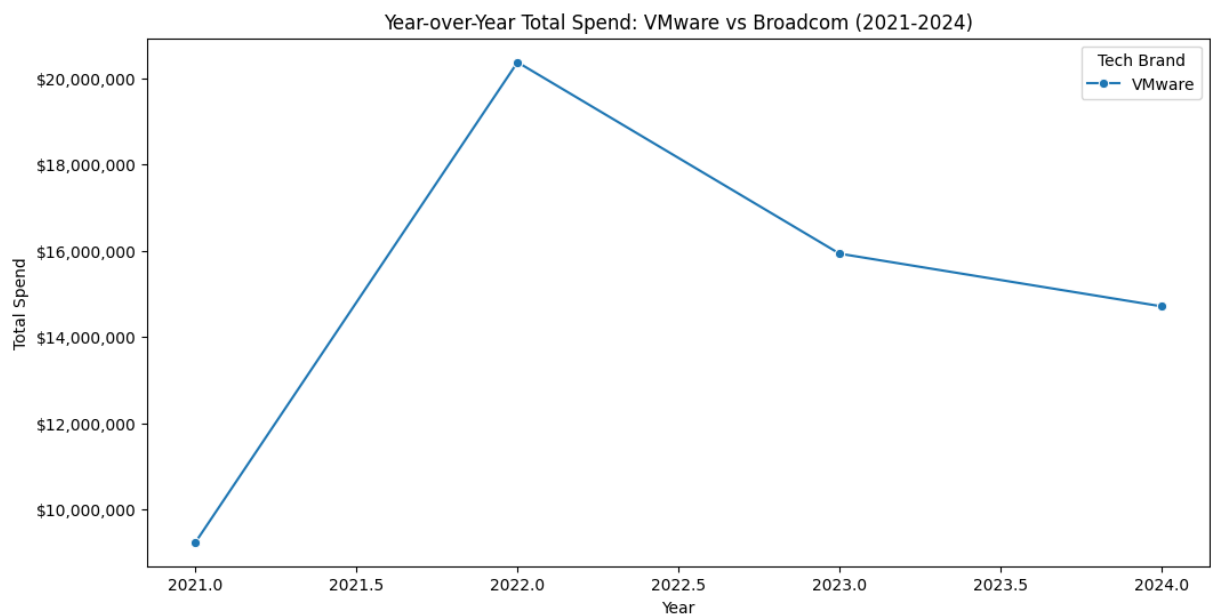


Figure 9. VMware by Broadcom SLED Procurement Spending Trend. A 2022 peak is followed by two years of declining expenditure, coinciding with the Broadcom acquisition.

VI. Business Value and Operational Impact

Prior to this project, the GovSpend platform’s native interface, while containing the underlying data, was effectively unusable at territory scale due to the limitations of its filtering and export tooling and the absence of any structured categorization for technology purchases. The processing pipeline and dashboard delivered through this project transform that raw resource into a fluid, self-service intelligence asset accessible to the sales organization.

For Nutanix Systems Engineers and account executives, the immediate operational benefit is the elimination of manual research time required to develop a customer technology profile prior to engagement. A user can now look up a specific agency in seconds and retrieve a structured view of its five-year IT procurement history, competitive install base, and channel relationships. Beyond individual account research, the dashboard enables territory-level strategic planning: identifying geographic

concentrations of specific competitor technologies, anticipating upcoming refresh cycles, and evaluating the depth of reseller relationships across the territory.

An additional dimension of business value is the acceleration of opportunity triage. If a prospect's procurement history reveals a hard disqualifier—such as a recently completed, long-term infrastructure investment with a competing vendor, or a budget pattern inconsistent with a near-term purchase—the sales team can reallocate their engagement effort earlier and more efficiently. In a market where public-sector sales cycles are often protracted and resource-intensive, the ability to prioritize opportunities with data-driven precision constitutes a meaningful productivity lever.

VII. Conclusion and Recommendations for Future Work

This project successfully demonstrated that publicly available government procurement data, when processed through a purpose-built NLP pipeline, can be transformed into structured, high-value market intelligence accessible at the point of sales engagement. The combination of a logistic regression gatekeeper model for IT classification and a custom spaCy NER pipeline for vendor and category extraction achieved sufficient accuracy—90% on vendor extraction against the Golden Dataset—to produce reliable intelligence outputs at scale. The resulting Power BI dashboard and trend analysis report represent practical, immediately deployable analytics tools for Nutanix's SLED pre-sales organization.

Several directions for future enhancement are recommended. First, the geographic scope of the dataset could be expanded beyond Georgia and Florida to encompass Nutanix's full southeastern SLED territory, substantially increasing the strategic utility of the tool. Second, a regular data refresh cadence—ideally aligned with Nutanix's Quarterly Business Review cycle—would ensure that the intelligence remains current as new procurement records become available. Third, continued expansion and refinement of the technology vendor entity dictionary could improve extraction coverage, particularly as the IT landscape evolves with new product families, acquisitions, and brand changes. Fourth, a "Multi-Pass" hybrid extraction approach—combining the current dictionary-based entity ruler with a generative language model for disambiguation of ambiguous or novel descriptions—could further improve performance on the most challenging line-item texts, though the compute infrastructure costs might outweigh the additional benefit. Finally, adoption of the dashboard within Nutanix Pre-Sales SE opportunity workflow could help accelerate the timeline from insight to sales execution.

References

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